Decoding ML

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Liberies

```r
library(dplyr)
library(tidyverse)
library(stringr)
library(naniar)
library(gridExtra)
library(grid)
library(mice)
library(dlookr)
```

```r
data <- read.csv("projectdata.csv")
names(data)
```

```
##  [1] "Study.ID"                      "Patient.ID"
##  [3] "Sample.ID"                     "Cancer.Type"
##  [5] "Cancer.Type.Detailed"          "Ethnicity"
##  [7] "Fraction.Genome.Altered"       "Sex"
##  [9] "Gene.Panel"                    "Metastatic.Site"
## [11] "MSI.Comment"                   "MSI.Score"
## [13] "MSI.Type"                      "Mutation.Count"
## [15] "Oncotree.Code"                 "Overall.Survival..Months."
## [17] "Overall.Survival.Status"       "Primary.Tumor.Site"
## [19] "Race"                          "Sample.Class"
## [21] "Number.of.Samples.Per.Patient" "Sample.coverage"
## [23] "Sample.Type"                   "Somatic.Status"
## [25] "Tumor.Purity"
```

```r
# Select the relevant columns
selected_data <- data[, c("Mutation.Count",
                          "Somatic.Status",
                          "Gene.Panel",
                          "Fraction.Genome.Altered",
                          "MSI.Type",
                          "Sex",
                          "Race",
                          "Ethnicity",
                          "Sample.Type",
                          "Overall.Survival.Status",
                          "Overall.Survival..Months.")]
```

I read the original dataset and create a new data set with only needed variables before cleaned the data!

Checking data structure

```
## 'data.frame':    3879 obs. of  11 variables:
##  $ Mutation.Count           : int  4 1 7 10 5 1 4 4 2 6 ...
##  $ Somatic.Status           : chr  "Matched" "Matched" "Matched" "Matched" ...
##  $ Gene.Panel               : chr  "IMPACT341" "IMPACT341" "IMPACT341" "IMPACT341" ...
##  $ Fraction.Genome.Altered  : num  0.278 0.315 0.35 0.404 0.429 ...
##  $ MSI.Type                 : chr  "Stable" "Indeterminate" "Stable" "Indeterminate" ...
##  $ Sex                      : chr  "FEMALE" "Female" "FEMALE" "FEMALE" ...
##  $ Race                     : chr  "WHITE" "White" "WHITE" "WHITE" ...
##  $ Ethnicity                : chr  "Non-Spanish; Non-Hispanic" "Non-Spanish; Non-Hispanic" "Non-Span
##  $ Sample.Type              : chr  "Primary" "Primary" "Metastasis" "Primary" ...
##  $ Overall.Survival.Status  : chr  "1:DECEASED" "0:LIVING" "1:DECEASED" "1:DECEASED" ...
##  $ Overall.Survival..Months.: num  3.55 132.66 13.68 13.31 29.16 ...
```

Checking unique objject in each variable

```r
# Function to show unique values for character columns
show_unique_characters <- function(data) {
  # Loop through each column in the dataset
  for (col_name in colnames(data)) {
    # Check if the column is of type character or factor
    if (is.character(data[[col_name]]) | is.factor(data[[col_name]])) {
      cat("\nUnique values in column:", col_name, "\n")
      print(unique(data[[col_name]]))  # Show unique values
    }
  }
}

# Call the function with your dataset
show_unique_characters(selected_data)
```

```
##
## Unique values in column: Somatic.Status
## [1] "Matched"   "Unmatched"
##
## Unique values in column: Gene.Panel
## [1] "IMPACT341" "IMPACT410" "IMPACT468" "IMPACT505"
##
## Unique values in column: MSI.Type
## [1] "Stable"        "Indeterminate" "Do not report" "Instable"
## [5] ""
##
## Unique values in column: Sex
## [1] "FEMALE"  "Female"  "Male"    "Unknown" "MALE"
##
## Unique values in column: Race
##  [1] "WHITE"                         "White"
```

```
##  [3] "ASIAN-FAR EAST/INDIAN SUBCONT" "I choose not to answer"
##  [5] "BLACK OR AFRICAN AMERICAN"     "PT REFUSED TO ANSWER"
##  [7] "Black or African American"     "OTHER"
##  [9] "Unknown"                       "Asian-Far East/Indian Subcont"
## [11] "UNKNOWN"                       "Chinese"
## [13] "Other"                         "NATIVE AMERICAN-AM IND/ALASKA"
## [15] "NO VALUE ENTERED"              "Middle Eastern"
## [17] "Caribbean Indian"
##
## Unique values in column: Ethnicity
##  [1] "Non-Spanish; Non-Hispanic"
##  [2] "Dominican Republic"
##  [3] "Unknown whether Spanish or not"
##  [4] "Spanish  NOS; Hispanic NOS, Latino NOS"
##  [5] "Unknown"
##  [6] "South/Central America (except Brazil)"
##  [7] "Puerto Rican"
##  [8] "Mexican (includes Chicano)"
##  [9] "Cuban"
## [10] "Spanish surname only"
##
## Unique values in column: Sample.Type
## [1] "Primary"          "Metastasis"       "Local Recurrence" "Unknown"
##
## Unique values in column: Overall.Survival.Status
## [1] "1:DECEASED" "0:LIVING"   ""
```

Dealing with “ ”

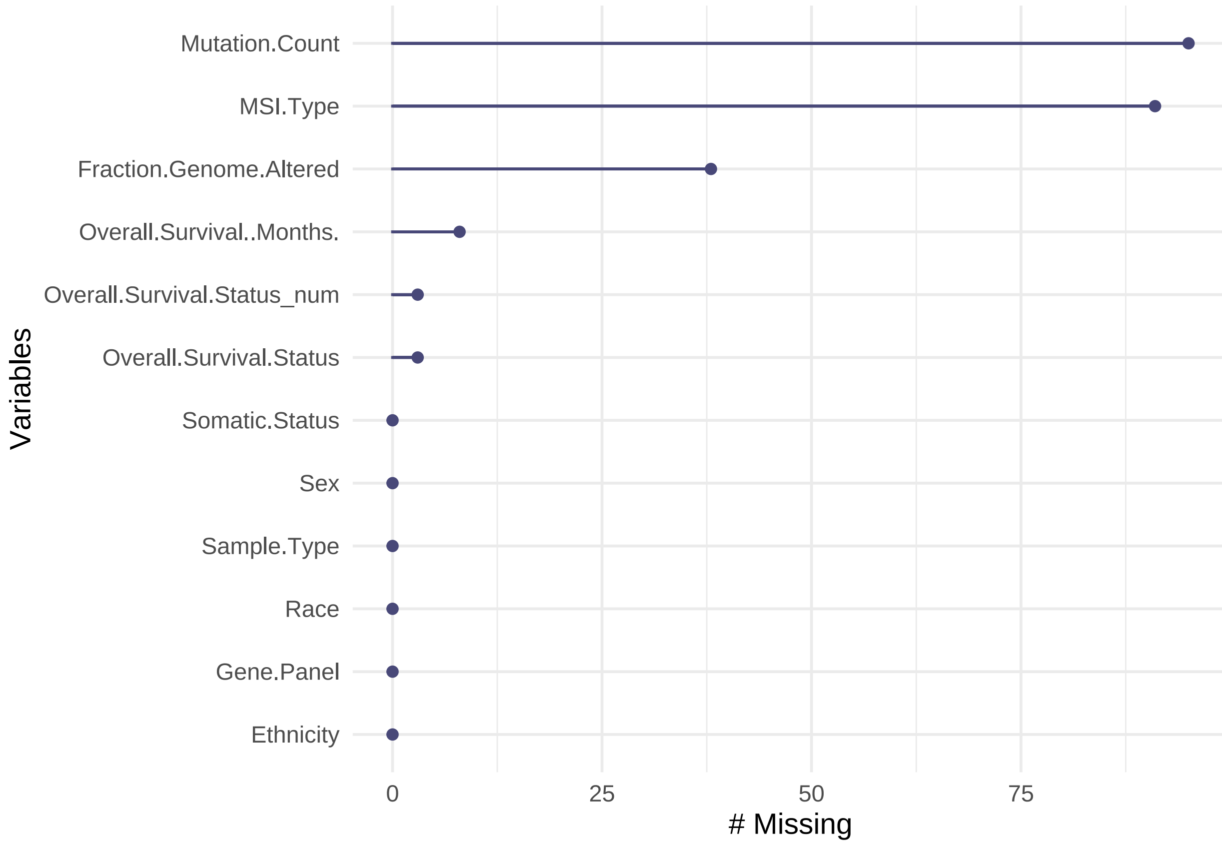
Correcting the unique object

```
selected_data<-selected_data %>%
  mutate(Sex = str_to_sentence(Sex),
         Race = str_to_sentence(Race))

selected_data %>%
  separate(Overall.Survival.Status, into = c("Overall.Survival.Status_num","Overall.Survival.Status")) %
  mutate(Overall.Survival.Status = str_to_sentence(Overall.Survival.Status),
         Gene.Panel = str_to_sentence(Gene.Panel)) -> selected_data
```

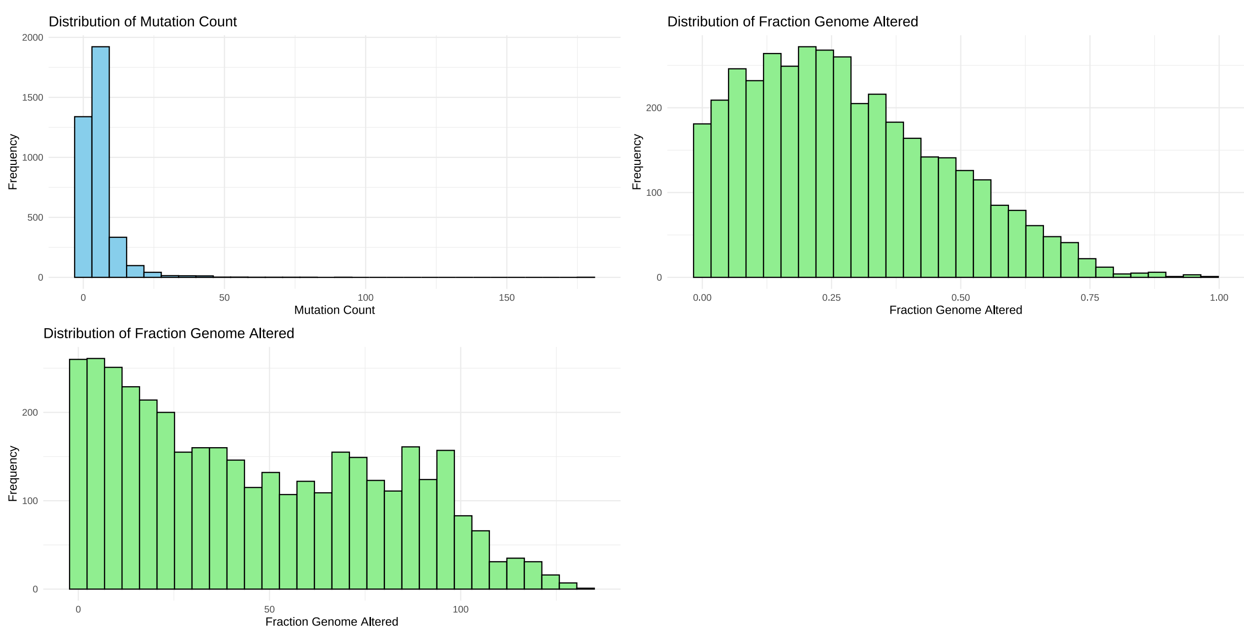
Checking the Missing variables

```
gg_miss_var(selected_data)
```



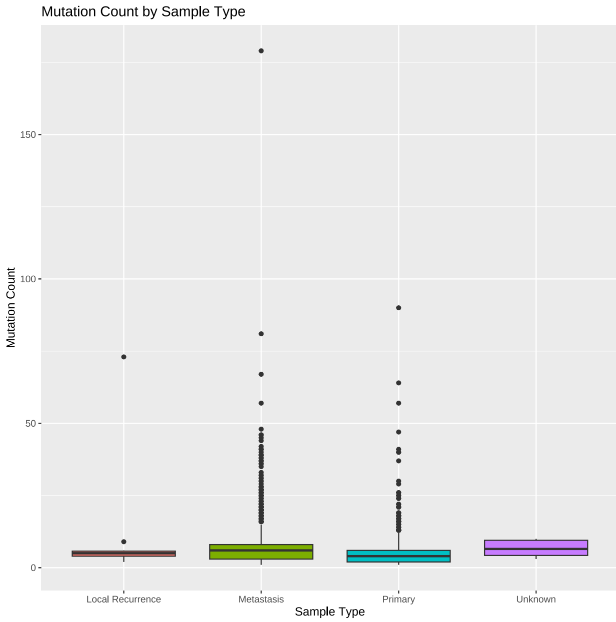
Distribution of Continous variables

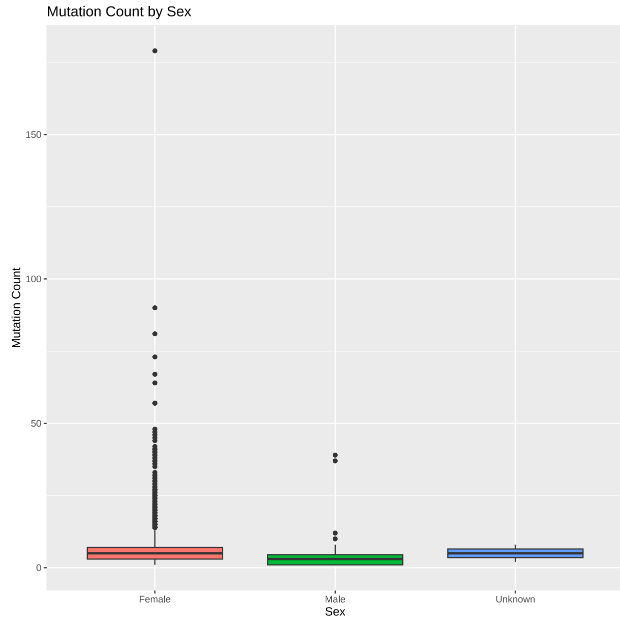
Histograms or Density Plots for Continuous Variables (e.g., Mutation Count, Fraction Genome Altered)



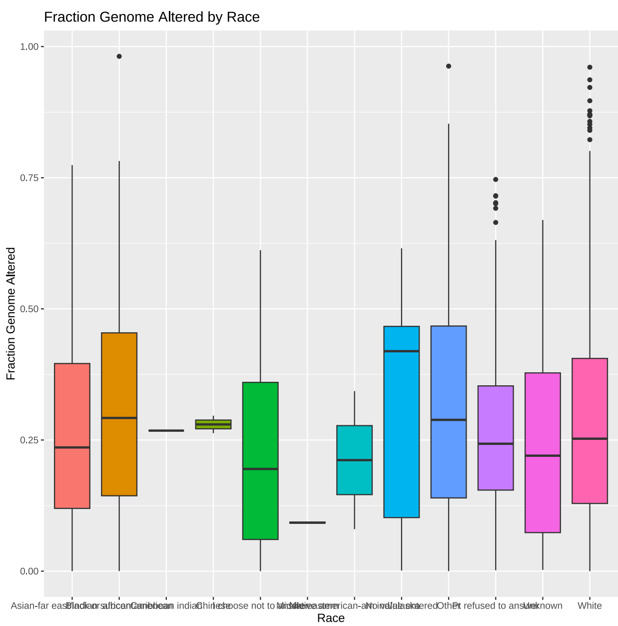
Boxplots for Genetic and Demographic Factors

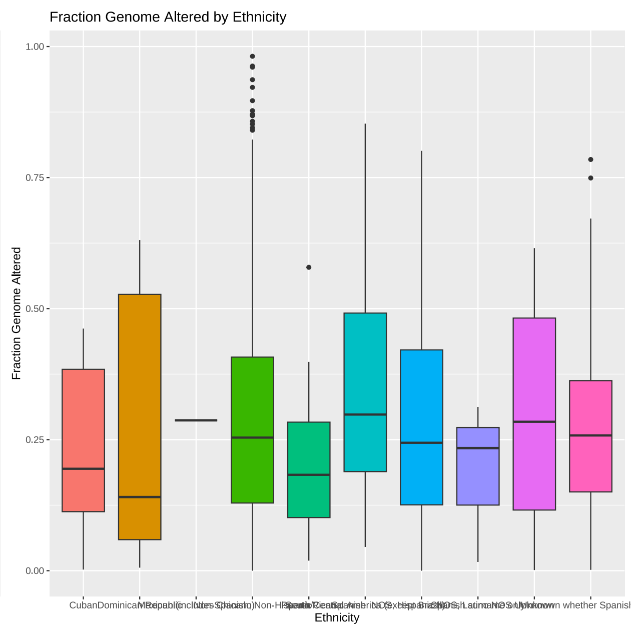
Boxplot of Mutation Count by Sample Type or Sex



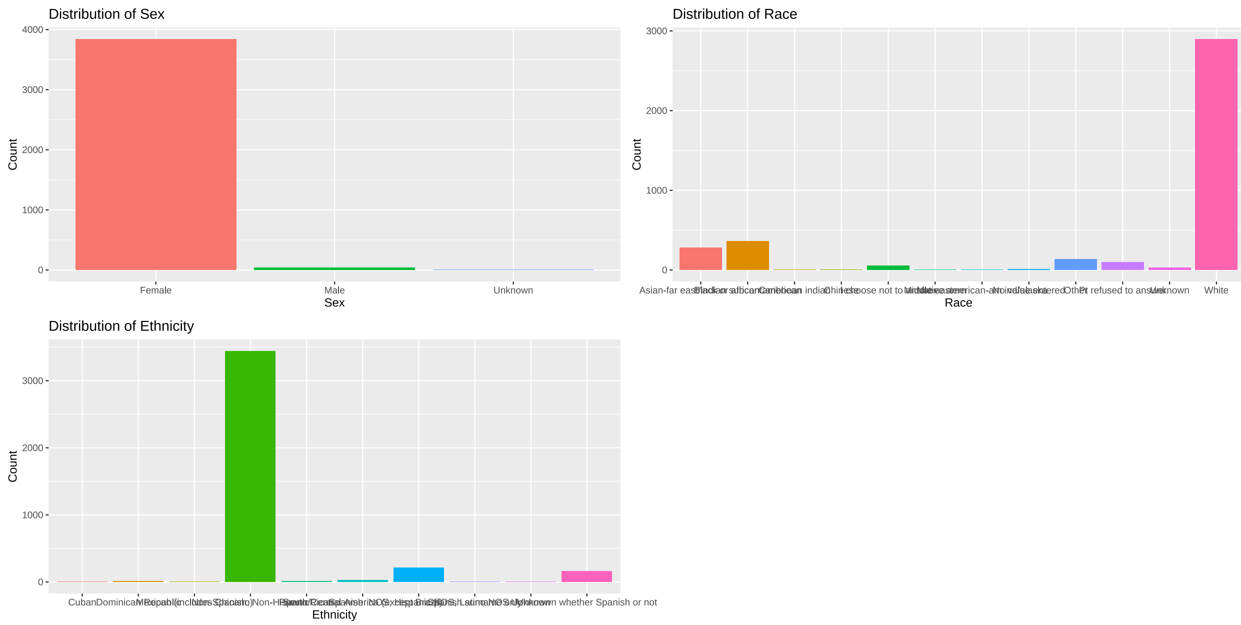


Boxplot of Fraction Genome Altered by Race or Ethnicity





Bar Plot for Sex, Race, and Ethnicity



Handling Missing and Unknown

```
##
##  iter imp variable
##   1   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   1   2  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   1   3  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   1   4  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   1   5  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   2   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   2   2  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   2   3  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   2   4  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   2   5  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   3   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   3   2  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   3   3  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   3   4  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   3   5  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   4   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
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##   4   3  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
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##   5   5  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
```

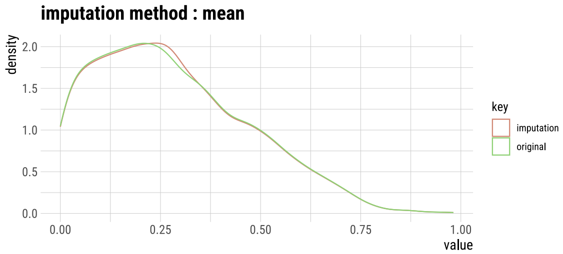

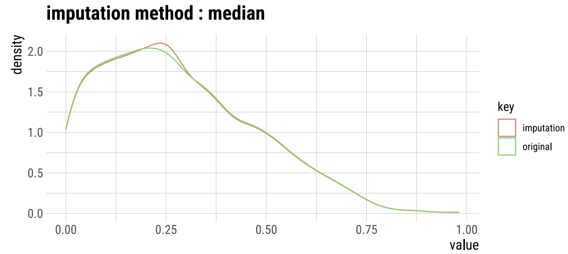

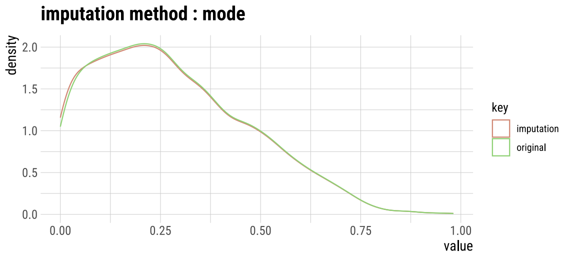

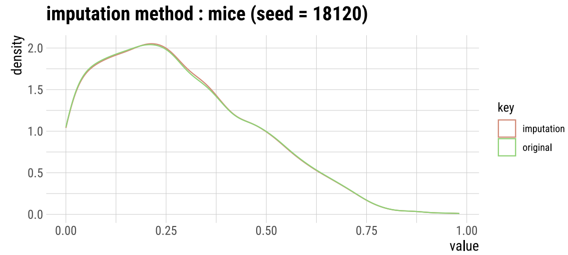

Look at the graph above it shows that mice imputation matches well than another methods

```
##
##   iter imp variable
##    1   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##    1   2  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
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```

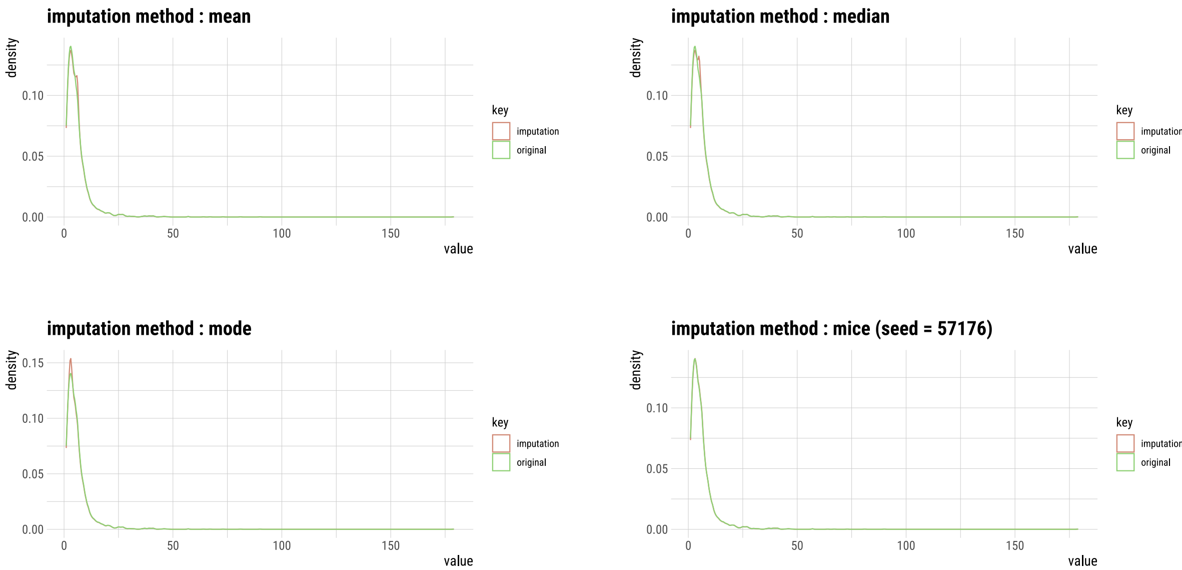


```
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##   iter imp variable
##    1   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##    1   2  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
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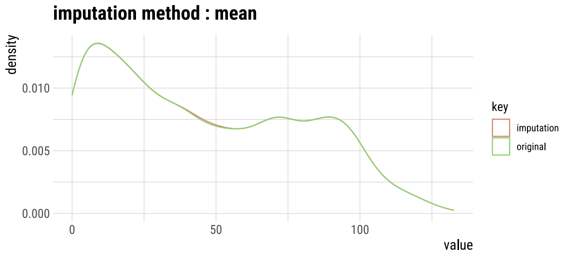

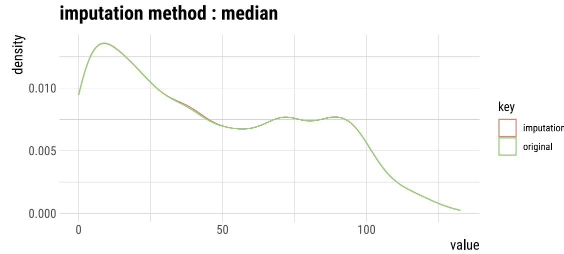

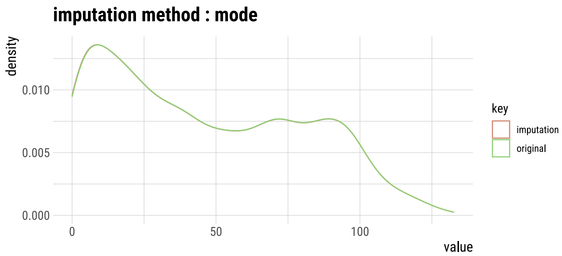

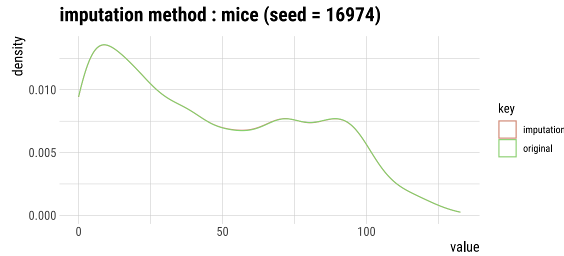


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##  iter imp variable
##   1   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##   1   2  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
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```

```
##    1   5  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##    2   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
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##    3   4  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##    3   5  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##    4   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##    4   2  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
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##    4   4  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
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##    2   1  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
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##    5   3  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##    5   4  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
##    5   5  Mutation.Count  Fraction.Genome.Altered  Overall.Survival..Months.
```

Function for imputing na for character type

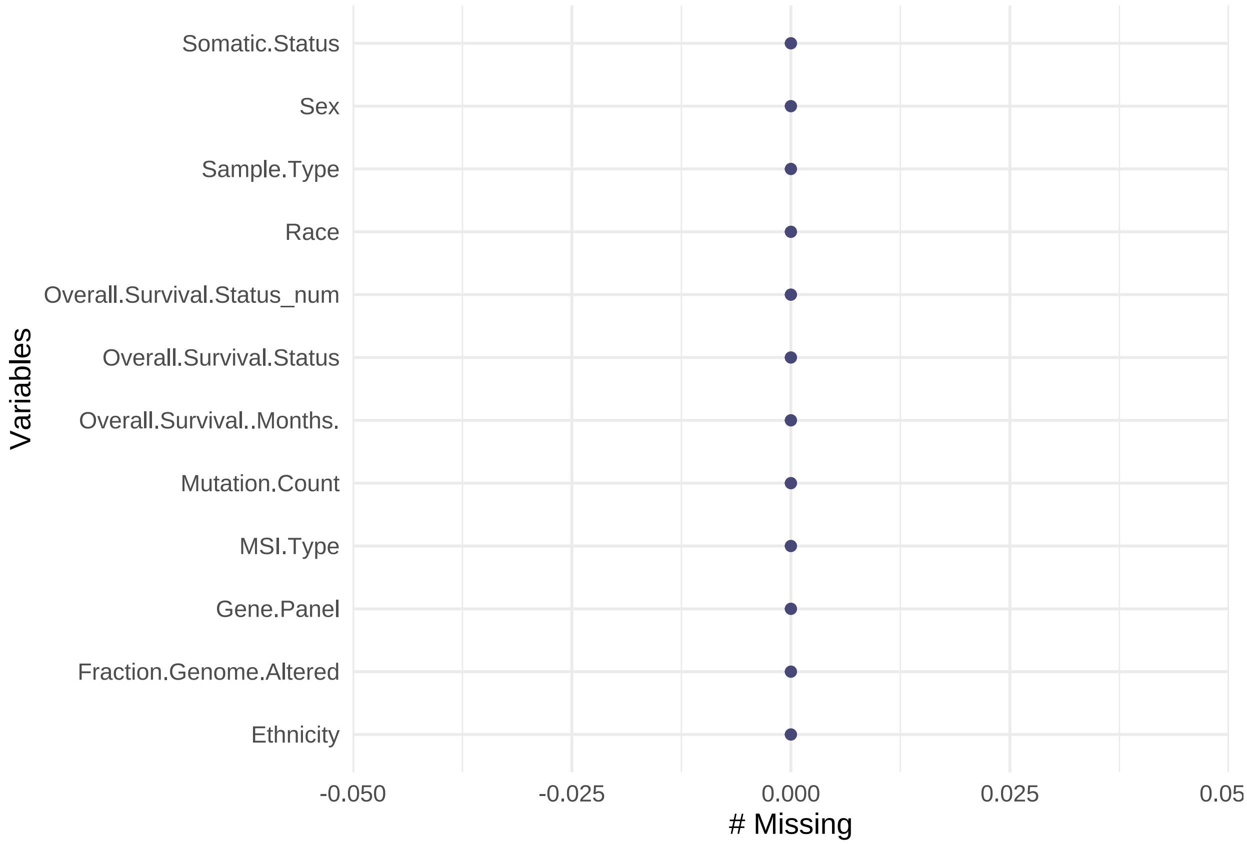
```r
# Function to impute missing values with the mode for character columns
impute_na_with_mode <- function(data) {
  # Loop through each column in the dataset
  data[] <- lapply(data, function(x) {
    # Only apply to character or factor columns
    if (is.character(x) | is.factor(x)) {
      # Calculate the mode (most frequent value)
      mode_value <- names(sort(table(x), decreasing = TRUE))[1]
      # Replace NAs with the mode value
      x[is.na(x)] <- mode_value
    }
    return(x)
  })
  return(data)
}

# Call the function with your dataset
selected_data <- impute_na_with_mode(selected_data)
```

I wrote a function that impute all missing values in character type data with mode

Confirmation Missing variables

```r
gg_miss_var(selected_data)
```



Droping all values with unknow

```
## # A tibble: 3 x 2
##   Sex         n
##   <chr>   <int>
## 1 Female   3840
## 2 Male       36
## 3 Unknown     3
```

```
## # A tibble: 12 x 2
##    Race                              n
##    <chr>                         <int>
##  1 Asian-far east/indian subcont   280
##  2 Black or african american       363
##  3 Caribbean indian                  1
##  4 Chinese                           2
##  5 I choose not to answer           55
##  6 Middle eastern                    2
##  7 Native american-am ind/alaska     2
##  8 No value entered                  9
##  9 Other                           138
## 10 Pt refused to answer            100
## 11 Unknown                          31
## 12 White                          2896
```

```
## # A tibble: 10 x 2
##    Ethnicity                                  n
##    <chr>                                  <int>
##  1 Cuban                                      6
##  2 Dominican Republic                        11
##  3 Mexican (includes Chicano)                 1
##  4 Non-Spanish; Non-Hispanic               3439
##  5 Puerto Rican                              11
##  6 South/Central America (except Brazil)     26
##  7 Spanish  NOS; Hispanic NOS, Latino NOS   216
##  8 Spanish surname only                       3
##  9 Unknown                                    7
## 10 Unknown whether Spanish or not           159
```

```
## # A tibble: 4 x 2
##   Sample.Type          n
##   <chr>            <int>
## 1 Local Recurrence    11
## 2 Metastasis        2050
## 3 Primary           1812
## 4 Unknown              6
```

```
selected_data %>%
  filter(Sex == "Unknown" | Race == "Unknown" | Ethnicity %in% c("Unknown whether Spanish or not","Unkn
```

```
## [1] 188  12
```

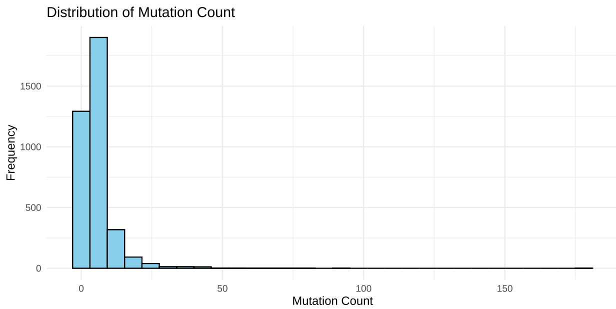
```
selected_data %>%
  filter(Sex != "Unknown" & Race != "Unknown" &
         !Ethnicity %in% c("Unknown whether Spanish or not", "Unknown") &
         Sample.Type != "Unknown") ->selected_data
```

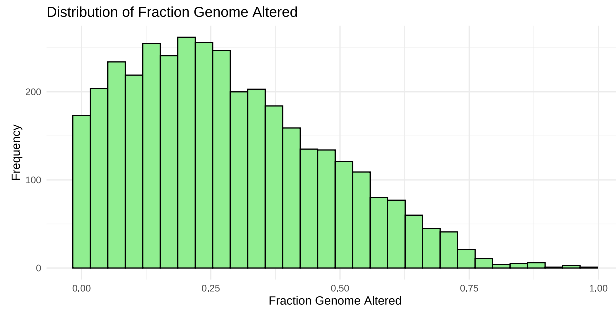
188 observation was drop because they have unkown

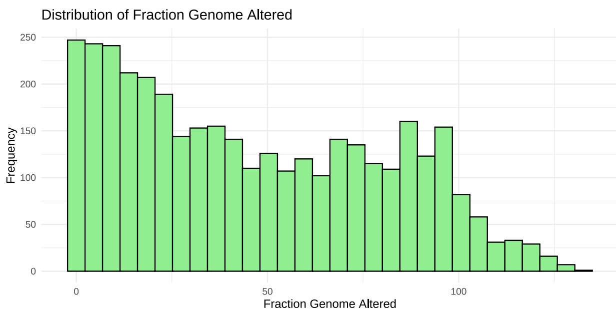
New Analysis After Data cleaning...

Distribution of Continous variables

Histograms or Density Plots for Continuous Variables (e.g., Mutation Count, Fraction Genome Altered)

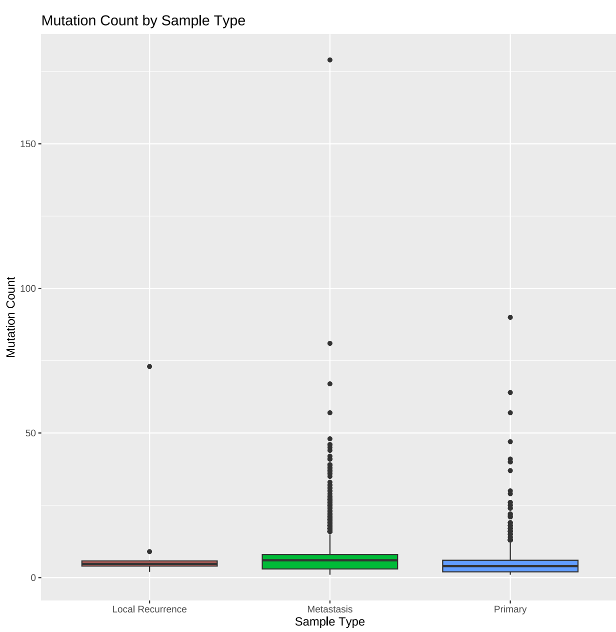


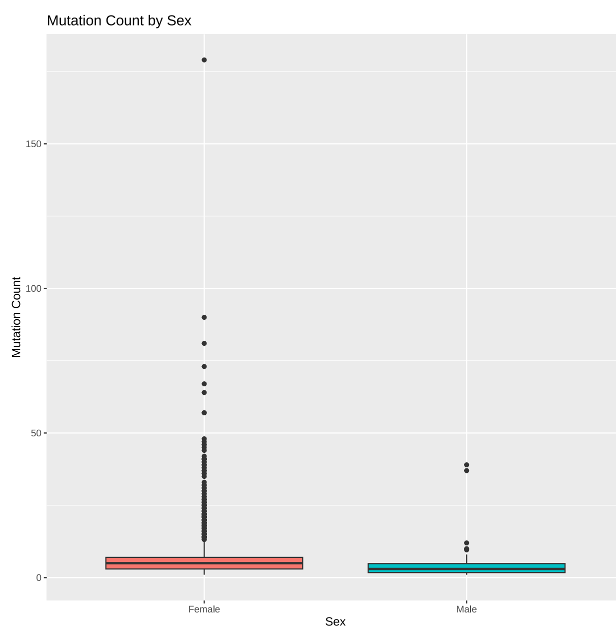




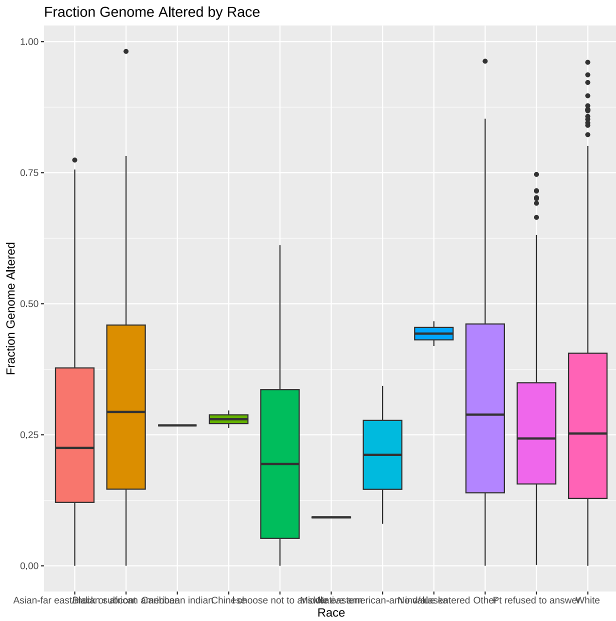
Boxplots for Genetic and Demographic Factors

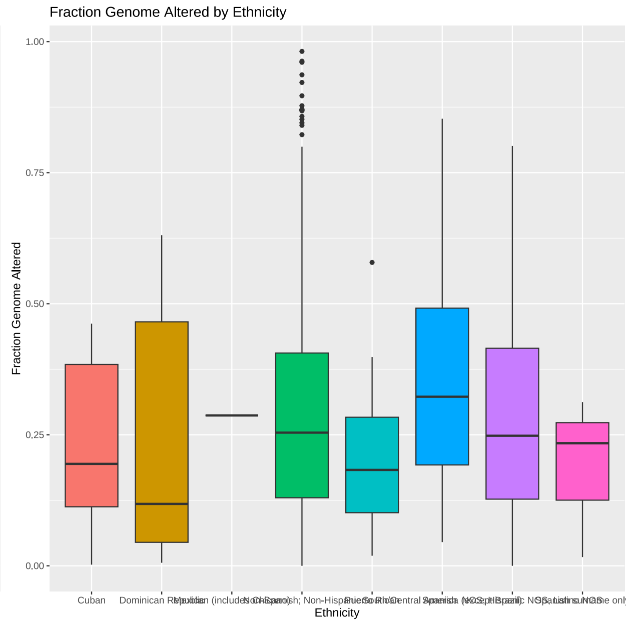
Boxplot of Mutation Count by Sample Type or Sex





Boxplot of Fraction Genome Altered by Race or Ethnicity





Bar Plot for Sex, Race, and Ethnicity